

GRADE 9 MATHEMATICS · STATISTICS & PROBABILITY

Probability

THE BIG IDEA

Independent events have no effect on the probability of the next event.
They **never** change the next probability.

INDEPENDENT VS. DEPENDENT

$$P(A) = \text{favourable} \div \text{total}$$

Independent: $P(A \text{ and } B) = P(A) \times P(B)$

Dependent: recalculate the total **after each draw**

**no memory. Past results
availability.**

WORKED EXAMPLE

Bag: 3 red, 5 blue — draw twice, no repla

$$P(\text{red on draw 1}) = 3/8$$

After removing 1 red: 2 red left, 7 total left

$$P(\text{red then red}) = 3/8 \times 2/7 = 3/28$$

⚠ WATCH OUT FOR

MISTAKE

Gambler's fallacy — "heads is due"

MISTAKE

Not updating the sample space (dependent events)

MISTAKE

cement

FIX

Every independent event starts fresh

FIX

Recalculate the total **and** favourable count after each draw

FIX



SELF-CHECK

A bag has 4 red and 6 green. Two drawn with
green)?

complete

without replacement. Can you find $P(\text{red then$